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Detailed speaker notes — predictive modelling

Slides 7–11 | target ≈2:00 (stretch 2:30)

Your run of slides

7. Modelling approach — multi-model search
8. MLflow tracking + registry
9. Results
10. Explainability (SHAP + LIME)
11. Fairness + four-type drift

Numbers cheat-sheet

- Mean target ≈**0.33** trips / 15-min slot
- Tuned val RMSE **0.994** dep / **0.976** arr
- Naive val RMSE 1.040 dep / 1.027 arr (≈**4% better**)
- Test R^2 **0.334** dep / **0.339** arr
- R^2 by tier (dep, test): **0.018 / 0.066 / 0.394**
- MLflow runs: **73** departure / **57** arrival

Hand-off in (from Louis): "Thanks, Louis. So we have clean, leakage-safe feature tables and a pipeline to train on. I own the modelling — how we pick a model, track it, and prove it's both accurate and responsible."

Slide 7 — Modelling: a multi-model search, auto-selected *(≈25–30s)*

"We deliberately did *not* hand-pick a model. We run a search. We start with two strong gradient-boosting baselines — LightGBM and XGBoost. On top of that we add **FLAML**, an AutoML layer that searches model families and configurations, and **Optuna**, which does Bayesian hyper-parameter tuning. *Every* trial is logged to MLflow. The pipeline then selects the best model purely by **validation RMSE** and promotes it automatically. We run this once per target — departures and arrivals — and, independently, both land on the same winner: an Optuna-tuned LightGBM, 'lgbm_optuna'. So the choice is automatic and reproducible, not a guess."

Hit these:

- Candidates (LightGBM + XGBoost) **+ FLAML AutoML + Optuna HPO**; every trial → MLflow.
- Selection is **by validation RMSE**, automatic; both targets independently pick `lgbm_optuna`.

Transition: *"Because every trial is tracked, we can stand behind that choice — here's the tracking."*

Slide 8 — MLflow tracking + registry *(≈20–25s)*

"All of that is fully traceable. We run an MLflow tracking server on EC2 with an S3 artifact store. The departure experiment has **73 runs**, arrival **57** — every candidate, FLAML and Optuna trial, with its parameters, metrics and the model itself. The single best run per target is registered in the model registry and promoted to the **production** alias — and that production model is exactly the artifact our apps load and serve. No retraining drift between what we evaluated and what we ship."

Hit these:

- Tracking server on **EC2 + S3**; experiments `bixi-demand-departure` / `-arrival`.
- Best run → **registered** → **production** alias = the exact artifact served.

Transition: *"So how good is that production model? Here are the numbers."*

Slide 9 — Results *(≈30–35s — your anchor slide)*

"Two things to read here. First, be honest about the metric: 15-minute, per-station demand is genuinely noisy — the average slot is about a third of a trip, and most slots are zero — so absolute R^2 is modest *by design*. What matters operationally is the error size, and that's well under **one trip per slot**: RMSE around 1.0, typical error about 0.6.

Second, it's a real model, not a baseline. Against a naive historical-average baseline we cut validation RMSE by about **4%** and add roughly six points of R^2. It's **stable across two unseen months** — May and October 2025 — so no overfitting. And the predictive power is concentrated exactly where decisions are made: on high-demand stations the R^2 is **0.39**, versus near zero on the quietest ones."

Hit these:

- Errors < **1 trip / slot** (RMSE ≈1.0, MAE ≈0.6); mean target only ≈0.33.
- **Beats naive** (≈4% RMSE, +6 pts R^2); **stable** on two unseen months.
- R^2 by tier (dep, test): low 0.018, medium 0.066, **high 0.394** — strong where it counts.

Transition: *"And we can explain* why *it predicts what it does."*

Slide 10 — Explainability: SHAP & LIME *(≈20–25s)*

"We explain both models with SHAP and LIME. The SHAP beeswarms tell a clean, sensible story: most of the signal comes from the **2024 historical baselines** — recent-lag and historical-average demand — and from the **cyclical time-of-day** features. Weather is a real but *secondary* driver. It's the same picture for departures and arrivals, which is reassuring — the model is leaning on stable demand structure, not noise. LIME lets us explain any single prediction on demand, which matters for operator trust."

Hit these:

- Top drivers: `baseline_prev_15min`, `hist_avg_demand`, cyclical time; weather secondary.
- Same story dep & arr → sensible behaviour; **LIME** for per-prediction explanations.

Transition: *"Explainability leads straight into responsible AI — fairness and drift."*

Slide 11 — Fairness & four-type drift *(≈25–30s)*

"Two responsible-AI checks. **Fairness:** we measure error parity — RMSE and MAE — across demand tiers and geography. As the tier numbers showed, accuracy concentrates in high-demand stations, and on the quietest stations R^2 is near zero. We don't hide that — we *surface* it so operators don't over-trust a quiet-station forecast.

Drift: we monitor all four types with Evidently — **feature, target, prediction and concept** drift. On the October-2025 data you can see feature drift is detected on a majority of columns, which is exactly the kind of seasonal shift the system is meant to catch. That's the hook for scheduled monitoring and retraining. That's the modelling story — over to Rui for what we do with it."

Hit these:

- Fairness = **error parity** across tiers/geography; weak on low-demand → flagged, not hidden.
- **All four** Evidently drift types (feature / target / prediction / concept), shown on Oct-2025.

Transition: *"Over to Rui — turning these forecasts into a rebalancing plan."*

Likely questions (be ready)

- **"Why is R^2 so low?"** — It's a 15-minute, per-station target: near-Poisson, mostly zeros, huge per-slot variance, so R^2 understates skill. We predict the conditional mean well (MAE ≈0.6 trip). It beats the naive baseline, is stable out-of-sample, and reaches R^2 0.39 on the busy stations where rebalancing decisions actually happen. Aggregate it to hourly/station-day and it looks much higher.
- **"Why LightGBM and not the FLAML pick / XGBoost?"** — We didn't decide; the search did. FLAML and Optuna both explored the space; `lgbm_optuna` had the lowest validation RMSE for *both* targets, so it was auto-selected and promoted.
- **"Isn't target encoding a leakage risk?"** — Encoders and the 2024 baselines are fit on **train only** (the baselines leave-one-out), then applied forward to May/Oct-2025. Strict temporal split, no peeking. (Sarah's slide 3.)
- **"What happens when drift fires?"** — Evidently flags it; that's the trigger for a scheduled re-run of the same pipeline to retrain and re-register a fresh production model — the roadmap item Rui mentions.
- **"Departures and arrivals — two models or one?"** — One shared pipeline, run twice (once per target). Same features, same selection logic; their difference is the rebalancing signal.

Timing: 5 slides in ≈2 min is brisk — Slide 9 (Results) is your anchor; if you're running long, compress Slides 8 and 10 to one sentence each. Keep the naive-baseline and tier-R^2 numbers — they carry the "it's a real, useful model" argument.